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Inventor(s)

Shin; Hoon et al.

### MEMORY CONTROLLER, MEMORY MODULE AND MEMORY SYSTEM

#### Abstract

A memory system is provided. The memory system includes: first memory devices; a second memory device; and a memory controller configured to control the plurality of first memory devices and the second memory device. Each of the first memory devices is configured to store a first data set including a first set of data bits in different bit positions, each of the bit positions corresponding to a burst order and a DQ. The second memory device is configured to store a second data set including parity bits for a plurality of first data sets, respectively stored in the plurality of first memory devices. A second set of data bits corresponding to a first parity bit of the parity bits are included in the plurality of first data sets, respectively, and at least two data bits of the second set of data bits have different bit positions.

**Inventors:** Shin; Hoon (Suwon-si, KR), Ahn; Sunggi (Suwon-si, KR), Kim; Sung-Rae (Suwon-si, KR), Kim; Wijong (Suwon-si, KR), Ryu; Yesin (Suwon-si, KR)

**Applicant:** SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)

**Family ID:** 1000008502631

**Assignee:** SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)

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## **Background/Summary**

### **CROSS-REFERENCE TO RELATED APPLICATIONS**

[0001] This application claims priority to Korean Patent Application No. 10-2024-0031613, filed on Mar. 5, 2024, in the Korean Intellectual Property Office, Korean Patent Application No. 10-2024-0054152, filed on Apr. 23, 2024, in the Korean Intellectual Property Office, and Korean Patent Application No. 10-2024-0162503, filed on Nov. 14, 2024, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

### **BACKGROUND**

[0002] The present disclosure relates to a memory controller, a memory module, and a memory system.

[0003] Memory devices are used to store data and may be categorized into volatile memory devices and nonvolatile memory devices. Volatile memory devices lose their stored data when their power supplies are interrupted. A dynamic random access memory (DRAM) is a type of volatile memory that is may be used in a variety of applications such as mobile systems, servers, or graphics devices.

[0004] A plurality of memory devices may be mounted on a single substrate and provided in the form of a memory module to achieve higher performance and higher capacity. A portion of the memory devices mounted on a memory module may store parity data to correct errors occurring in data stored in other memory devices.

### **SUMMARY**

[0005] One or more example embodiments provide a memory system performing an error correction operation with improved performance.

[0006] According to an aspect of an example embodiment, a memory system includes: a plurality of first memory devices; a second memory device; and a memory controller configured to control the plurality of first memory devices and the second memory device. Each of the plurality of first memory devices is configured to store a first data set including a first set of data bits in different bit positions, each of the bit positions corresponding to a burst order and a DQ. The second memory device is configured to store a second data set including parity bits for a plurality of first data sets, respectively stored in the plurality of first memory devices. A second set of data bits corresponding to a first parity bit of the parity bits are included in the plurality of first data sets, respectively, and at least two data bits of the second set of data bits have different bit positions.

[0007] According to another aspect of an example embodiment, a memory module includes: a plurality of first memory devices; and a second memory device. Each of the plurality of first memory devices is configured to store a first data set including a first set of data bits having bit positions, each of the bit positions corresponding to a burst order and a DQ. The second memory device is configured to store a second data set including first parity bits for a plurality of first data sets, respectively stored in the plurality of first memory devices. A second set of data bits corresponding to one parity bit of the first parity bits are included in the plurality of first data sets, respectively, and at least two data bits of the second set of data bits have different bit positions.

[0008] According to another aspect of an example embodiment, a method of operating a memory controller to control a memory module including a plurality of first memory devices and a second

memory device, is provided. The method includes: storing a plurality of first data sets in the plurality of first memory devices, respectively; and storing a second data set, including parity bits for the plurality of first data sets, in the second memory device. Each of the plurality of first data sets includes a first set of data bits having bit positions, each of the bit positions corresponding to a burst order and a DQ. A second set of data bits corresponding to a first parity bit of the parity bits are included in the plurality of first data sets, respectively, and bit positions of at least two data bits of the second set of data bits are different from each other.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

[0009] The above and other aspects and features of the present disclosure will be more clearly understood from the following description of example embodiments, taken in conjunction with the accompanying drawings, in which

[0010] FIG. 1 is a block diagram of an electronic system according to an example embodiment;

[0011] FIG. 2A is a diagrams illustrating bit positions of bits included in a data set according to an example embodiment;

[0012] FIG. 2B is a diagram illustrating related parity mapping;

[0013] FIG. 2C is a diagram illustrating parity mapping according to an example embodiment;

[0014] FIG. 3 is a diagram illustrating a bank according to an example embodiment;

[0015] FIG. 4 is a block diagram of a memory device according to an example embodiment;

[0016] FIG. 5 is a block diagram of a memory controller according to an example embodiment;

[0017] FIG. 6 is a flowchart illustrating a method of operating a memory system according to an example embodiment;

[0018] FIGS. 7A and 7B are diagrams illustrating a method of operating a memory system according to an example embodiment;

[0019] FIG. 8 is a flowchart illustrating a method of operating a memory system according to an example embodiment;

[0020] FIGS. 9A, 9B, 9C and 9D are diagrams illustrating a method of operating a memory system according to an example embodiment;

[0021] FIGS. 10A and 10B are diagrams illustrating limitations of an error correction operation during related parity mapping;

[0022] FIGS. 11A, 11B, 11C and 11D are diagrams illustrating a method of operating a memory system according to an example embodiment; and

[0023] FIG. 12 is a block diagram of an electronic system according to an example embodiment.

### DETAILED DESCRIPTION

[0024] Hereinafter, example embodiments will be described with reference to the accompanying drawings.

[0025] FIG. 1 is a block diagram illustrating the configuration of an electronic system according to an example embodiment.

[0026] Referring to FIG. 1, an electronic system 10 may include a host 15 and a memory system 20. The memory system 20 may include a memory controller 100 and a memory module 200. The memory module 200 may include a plurality of memory devices 200p and 200d\_1 to 200d\_n. The plurality of memory devices 200p and 200d\_1 to 200d\_n may include a plurality of data memory devices 200d\_1 to 200d\_n and a single parity memory device 200p.

[0027] Each of the plurality of memory devices 200p and 200d\_1 to 200d\_n may input or output bits in units of prefetch sizes. Hereinafter, a set of bits stored in a memory device through a single write operation or output from a memory device through a single read operation, corresponding to a prefetch size, is referred to as a “data set.”

[0028] The prefetch size is determined depending on the number of DQs (i.e., the data bus width or the number of data pins) used by each memory device and a burst length, so that bits included in a data set may have bit positions defined by a burst order and DQs.

[0029] According to example embodiments, each of the plurality of data memory devices **200d\_1** to **200d\_n** may store a first data set including data bits. The parity memory device **200p** may store a second data set including parity bits for a plurality of first data sets, respectively stored in the plurality of data memory devices **200d\_1** to **200d\_n**.

[0030] Data bits corresponding to one of the parity bits included in the second data set are included in different first data sets and may have different bit positions.

[0031] As described above, parity bits mapped to data bits having different bit positions in each of the first data sets may be stored in the parity memory device **200p**, allowing the memory system **20** to perform an error correction operation with improved performance.

[0032] A detailed description will now be provided with reference to FIG. 1. The host **15** may communicate with the memory system **20** using an interface protocol such as peripheral component interconnect-express (PCI-E), advanced technology attachment (ATA), serial ATA (SATA), parallel ATA (PATA), or a serial attached SCSI (SAS). The interface protocols between the host **15** and the memory system **20** are not limited to the above-mentioned examples and may be one of other interface protocols such as a universal serial bus (USB) protocol, an MMC protocol, an enhanced small disk interface (ESDI) protocol, or an integrated drive electronics (IDE) protocol.

[0033] The memory controller **100** may control the overall operation of the memory system **20**. The memory controller **100** may control the overall data exchange between the host **15** and the memory module **2000**. For example, the memory controller **100** may control the memory devices **200p** and **200d\_1** to **200d\_n** to write or read data in response to a request from the host **15**. Also, the memory controller **100** may control the operation of the memory devices **200p** and **200d\_1** to **200d\_n** by applying commands and/or addresses to control the memory devices **200p** and **200d\_1** to **200d\_n**.

[0034] According to an example embodiment, the memory controller **100** may control the plurality of first data sets to be stored in the plurality of data memory devices **200d\_1** to **200d\_n**, respectively. Also, the memory controller **100** may control the second data set to be set (i.e., stored) in the parity memory device **200p**. Each of the parity bits included in the second data set may have a value based on an XOR operation on corresponding data bits. For example, each of the parity bits may be even XOR parity, but example embodiments are not limited thereto. In some example embodiments, each of the parity bits may be odd XOR parity.

[0035] The memory controller **100** may receive the plurality of first data sets and the second data set from the plurality of data memory devices **200d\_1** to **200d\_n** and the parity memory device **200p**, and may perform an error correction operation on the received data sets.

[0036] For example, when data is requested by the host **15**, the memory controller **100** may apply a read command to the plurality of data memory devices **200d\_1** to **200d\_n** and the parity memory device **200p** to receive the plurality of first data sets and the second data set. Accordingly, the memory controller **100** may perform a parity check on the plurality of first data sets and the second data set to determine candidate error bit positions for each of the plurality of first data sets and the second data set.

[0037] The memory controller **100** may obtain first flags corresponding to each of the plurality of first data sets and the second data set. Each of the first flags may indicate a result of the error correction operation performed on each of the memory devices **200p** and **200d\_1** to **200d\_n**. For example, the first flags may include results of a plurality of error correction operations, each performed on the plurality of first data sets in the plurality of data memory devices **200d\_1** to **200d\_n**. Also, the first flags may include a result of the error correction operation performed on the second data set in the parity memory device **200p**. Accordingly, the memory controller **100** may determine an error data set including error bits, among the plurality of first data sets and the second

data set, based on the received first flags.

[0038] According to an example embodiment, the memory controller **100** may determine an error data set when the first flags have specific values. Even when the first flags do not have the specific values, the memory controller **100** may perform an iteration operation to again determine an error data set, which will be described later. Accordingly, there may be an increased likelihood of detecting an error data set. As described above, this is because parity bits are mapped to data bits having different bit positions in each of the first data sets and are stored in the parity memory device **200p**.

[0039] Accordingly, the memory controller **100** may flip bits corresponding to candidate error bit positions, among bits included in an error data set, to perform an error correction operation.

[0040] According to an example embodiment, the memory controller **100** may include a system error correction code (ECC) circuit **120**. The system ECC circuit **120** may perform operations related to an error correction operation, among the operations of the memory controller **100**. For example, the system ECC circuit **120** may perform an operation of mapping data bits and parity bits by varying bit positions, an operation of performing a parity check to determine candidate error bit positions, an operation of determining an error data set based on flags, and an operation of flipping bits corresponding to candidate error bit positions among the bits included in the error data set, but example embodiments are not limited thereto.

[0041] Each of the plurality of memory devices **200p** and **200d\_1** to **200d\_n** may receive a data set from the memory controller **100** and store the received data set. Also, each of the plurality of memory devices **200p** and **200d\_1** to **200d\_n** may read the stored data set in response to a request from the memory controller **100** and transmit the read data set to the memory controller **100**.

[0042] According to an example embodiment, the memory module **2000** may be a dual in-line memory module (DIMM). For example, the memory module **2000** may be a registered DIMM (RDIMM), a load reduced DIMM (LRDIMM), an unbuffered DIMM (UDIMM), a fully buffered DIMM (FB-DIMM), or a small outline DIMM (SO-DIMM). However, example embodiments are not limited thereto and the memory module **2000** may also be another memory module, such as a single in-line memory module (SIMM).

[0043] According to an example embodiment, each of the plurality of memory devices **200p** and **200d\_1** to **200d\_n** may be a memory device including volatile memory cells. For example, the memory device **200** may be various dynamic random access memory (DRAM) devices such as a double data rate synchronous DRAM (DDR SDRAM) device, a DDR2 SDRAM device, a DDR3 SDRAM device, a DDR4 SDRAM device, a DDR5 SDRAM device, a DDR6 SDRAM device, a low power double data rate (LPDDR) SDRAM device, an LPDDR2 SDRAM device, an LPDDR3 SDRAM device, an LPDDR4 SDRAM device, an LPDDR4X SDRAM device, an LPDDR5 SDRAM device, a graphics double data rate synchronous graphics random access memory (GDDR SGRAM) device, a GDDR2 SGRAM device, a GDDR3 SGRAM device, a GDDR4 SGRAM device, a GDDR5 SGRAM device, or a GDDR6 SGRAM device.

[0044] According to an example embodiment, each of the plurality of memory devices **200p** and **200d\_1** to **200d\_n** may be a memory device in which DRAM dies are stacked, such as a high bandwidth memory (HBM), an HBM2, or an HBM3.

[0045] According to an example embodiment, each of the plurality of memory devices **200p** and **200d\_1** to **200d\_n** may be an SRAM device, a NAND flash memory device, a NOR flash memory device, an RRAM device, an FRAM device, a PRAM device, a TRAM device, or an MRAM device.

[0046] According to an example embodiment, the plurality of data memory devices **200d\_1** to **200d\_n** may provide data corresponding to a cache line size. Therefore, the number of the plurality of data memory devices **200d\_1** to **200d\_n** may be determined depending on a prefetch size of each of the plurality of data memory devices **200d\_1** to **200d\_n**. For example, when the cache line size is 64 bytes (=512 bits) and the prefetch size of each of the plurality of data memory devices **200d\_1**

to **200d\_n** is 128 bits, the number of the plurality of data memory devices **200d\_1** to **200d\_n** may be four. When the cache line size is 64 bytes (=512 bits) and the prefetch size of each of the plurality of data memory devices **200d\_1** to **200d\_n** is 256 bits, the number of the plurality of data memory devices **200d\_1** to **200d\_n** may be two.

[0047] According to an example embodiment, the plurality of memory devices **200p** and **200d\_1** to **200d\_n** may include on-die ECC circuits **350p** and **350d\_1** to **350d\_n**, respectively, generating the above-mentioned first flag. For example, each of the on-die ECC circuits **350p** and **350d\_1** to **350d\_n** may perform an error correction operation on a data set input or output through the corresponding memory device using a parity check matrix (or H-matrix) and generate a first flag regarding a result of the error correction operation. The first flag may indicate a non-uncorrectable error (non-UE) when it is determined that there is no error or that the error has been corrected, or an uncorrectable error (UE) when it is determined that an uncorrectable error has occurred. The generated first flag may be provided to the memory controller **100** based on a request of the memory controller **100**.

[0048] According to an example embodiment, each of the plurality of data memory devices **200d\_1** to **200d\_n** may be referred to as a data chip. In addition, the parity memory device **200p** may be referred to as a module ECC chip, a system ECC chip, or the like, but example embodiments are not limited thereto.

[0049] As described above, the memory system **20** according to one or more example embodiments may perform an error correction operation with improved performance.

[0050] FIG. 2A is a diagrams illustrating bit positions of bits included in a data set according to an example embodiment. In FIG. 2A, for ease of description, an example is provided in which a prefetch size of a data set is 16 bits.

[0051] As illustrated in FIG. 2A, a memory device using a data set **30** has a burst length (BL) of 4 and uses four DQs. The data set may include 16 bits, respectively corresponding to 16 bit positions defined by a burst order BL0, BL1, BL2, and BL3 and DQs DQ0, DQ1, DQ2, and DQ3.

[0052] Similarly, bits included in a data set having another prefetch size may also have bit positions defined by the burst order and DQs.

[0053] FIG. 2B is a diagram illustrating related parity mapping. FIG. 2B illustrates first data sets Data 1 and Data 2 stored in two data memory devices having a prefetch size of 16 bits and a second data set parity stored in a parity memory device.

[0054] Referring to FIG. 2B, in the case of related technology, a parity bit for data bits having the same bit position in the first data sets Data 1 Data 2 may be stored at the same bit position in the second data set Parity.

[0055] For example, a parity bit “0” for data bits “11” having bit positions BL0 and DQ0 in Data 1 and Data 2 may be stored at the bit positions BL0 and DQ0 in the Parity. Also, for example, a parity bit “1” for data bits “01” having bit position BL2 and DQ3 in Data 1 and Data 2 may be stored at bit positions BL2 and DQ3 in the Parity. This is equally applied to the remaining data bits and parity bits.

[0056] FIG. 2C is a diagram illustrating parity mapping according to an example embodiment. FIG. 2C illustrates first data sets Data 1 and Data 2 stored in two data memory devices having a prefetch size of 16 bits and a second data set Parity stored in a parity memory device.

[0057] According to an example embodiment, at least two data bits among data bits corresponding to one of the parity bits included in the second data set Parity may have different bit positions within corresponding at least two first data sets, respectively.

[0058] Referring to FIG. 2C, a parity bit “0” for a data bit “1” having bit positions BL0 and DQ0 in Data 1 and a data bit “1” having bit positions BL0 and DQ3 in Data 2 may be stored at the bit positions BL0 and DQ2 in the parity. Also, for example, a parity bit “1” for a data bit “0” having bit positions BL2 and DQ3 in Data 1 and a data bit “1” having bit positions BL2 and DQ2 in Data 2 may be stored at the bit position BL2 and DQ1 in the parity. As described above, according to an

example embodiment, each data bit and a corresponding parity bit may be mapped to have different DQ positions. As illustrated in FIG. 2C, DQ0 of Data 1, DQ3 of Data 2, and DQ2 in the parity are mapped to each other, DQ1 of Data 1, DQ0 of Data 2, and DQ3 in the parity are mapped to each other, DQ2 of Data 1, DQ1 of Data 2, and DQ0 in the parity are mapped to each other, and DQ3 of Data 1, DQ2 of Data 2, and DQ1 in the parity are mapped to each other.

[0059] However, example embodiments are not limited thereto, and each data bit and a corresponding parity bit may be mapped to have different BL positions, or each data bit and a corresponding parity bit may be mapped to have different BL positions and different DQ positions.

[0060] FIG. 3 is a diagram illustrating a bank according to an example embodiment. A bank 300 of FIG. 3 may be one of the plurality of banks included in each of the plurality of memory devices 200p and 200d\_1 to 200d\_n of FIG. 1.

[0061] Referring to FIG. 3, the bank 300 may include a bank array 310\_1, a row decoder 260\_1, and a column decoder 270\_1. The bank array 310\_1 may include a plurality of memory cells. The plurality of memory cells may be electrically connected to a plurality of wordlines WL1 to WLj and a plurality of column select lines CSL1 to CSLi.

[0062] The bank array 310\_1 may include a normal cell area NA, in which a data set is stored, and a parity cell area PA in which a parity bit for the data set stored in the normal cell area NA is stored.

[0063] The row decoder 260\_1 may activate one of the plurality of rows ROW1 to ROWj in response to a row address RA. For example, the plurality of rows ROW1 to ROWj may correspond to the plurality of wordlines WL1 to WLj, respectively.

[0064] The column decoder 270\_1 may activate one of the plurality of column select lines CSL1 to COLi in response to a column address CA.

[0065] For example, when the bank 300 is included in data memory devices 200d\_1 to 200d\_n, the first data set including data bits may be stored in the normal cell area NA, and the parity bit for the first data set may be stored in the parity cell area PA.

[0066] When the bank 300 is included in the parity memory device 200p, a second data set including parity bits for the plurality of first data sets, respectively stored in the plurality of data memory devices 200d\_1 to 200d\_n, may be stored in the normal cell area NA, and parity bits for the second data set may be stored in the parity cell area PA.

[0067] FIG. 4 is a block diagram of a memory device according to an example embodiment. A memory device 200 of FIG. 4 may correspond to one of the plurality of memory devices 200p and 200d\_1 to 200d\_n of FIG. 1.

[0068] Referring to FIG. 4, the memory device 200 may include a control logic circuit 210, an address register 220, a bank control circuit 230, a refresh control circuit 400, a row address multiplexer 240, a column address latch 250, a row decoder group 260, a column decoder group 270, a memory cell array 310, a sense amplifier group 285, an input/output gating circuit 290, an on-die ECC circuit 350, and a data input/output buffer 320.

[0069] The memory cell array 310 may include a plurality of bank arrays 310\_1 to 310\_n. Each of the plurality of bank arrays 310\_1 to 310\_n may include a plurality of memory cells. For example, each of the plurality of memory cells may be formed at an intersection of a corresponding wordline and a corresponding bitline. Each of the plurality of bank arrays 310\_1 to 310\_n may correspond to the bank array 310\_1 of FIG. 3, but example embodiments are not limited thereto.

[0070] The row decoder group 260 may include a plurality of row decoders 260\_1 to 260\_n. Each of the plurality of row decoders 260\_1 to 260\_n may be connected to a corresponding bank array among the plurality of bank arrays 310\_1 to 310\_n.

[0071] The sense amplifier group 285 may include a plurality of sense amplifiers 285\_1 to 285\_n. Each of the plurality of sense amplifiers 285\_1 to 285\_n may be connected to a corresponding bank array among the plurality of bank arrays 310\_1 to 310\_n.

[0072] The column decoder group 270 may include a plurality of column decoders 270\_1 to 270\_n. Each of the plurality of column decoders 270\_1 to 270\_n may be connected to a corresponding

bank array of the plurality of bank arrays **310\_1** to **310\_n** through column select lines.

[0073] The address register **220** may receive an address ADDR including a bank address BANK\_ADDR, a row address ROW\_ADDR, and a column address COL\_ADDR from the memory controller **100**. The address register **220** may provide the received bank address BANK\_ADDR to the bank control circuit **230**, the received row address ROW\_ADDR to the row address multiplexer **240**, and the received column address COL\_ADDR to the column address latch **250**.

[0074] The bank control circuit **230** may generate bank control signals in response to the bank address BANK\_ADDR. For example, the row decoder corresponding to a bank address BANK\_ADDR, among the plurality of row decoders **260\_1** to **260\_n**, may be activated in response to the bank control signals. The column decoder corresponding to a bank address BANK\_ADDR, among the plurality of column decoders **270\_1** to **270\_n**, may be activated in response to the bank control signals.

[0075] The row address multiplexer **240** may receive a row address ROW\_ADDR from the address register **220** and a refresh row address REF\_ADDR from the refresh control circuit **400**. The row address multiplexer **240** may selectively output the row address ROW\_ADDR or the refresh row address REF\_ADDR as a row address RA. The row address RA, output from the row address multiplexer **240**, may be applied to each of the plurality of row decoders **260\_1** to **260\_n**.

[0076] The refresh control circuit **400** may sequentially increment or decrement the refresh row address REF\_ADDR in response to refresh signals from the control logic circuit **210** in a normal refresh mode.

[0077] The refresh control circuit **400** may receive a hammer address HADDR in a hammer refresh mode. The refresh control circuit **400** may output addresses of wordlines adjacent to an intensively accessed wordline as the refresh row address REF\_ADDR based on the hammer address HADDR.

[0078] A row decoder selected by the bank control circuit **230**, among the plurality of row decoders **260\_1** to **260\_n**, may activate a wordline corresponding to a row address RA output from the row address multiplexer **240**. For example, the selected row decoder may apply a wordline driving voltage to a wordline corresponding to a row address.

[0079] The column address latch **250** may receive a column address COL\_ADDR from the address register **220** and temporarily store the received column address COL\_ADDR.

[0080] According to an example embodiment, the memory device **200** may perform a burst operation in burst mode. The burst operation may refer to an operation of writing or reading a large amount of data by sequentially decrementing or incrementing the address from an initial address received by the memory device **200** from the memory controller **100**. A basic unit of such a burst operation may be a burst length. For example, the burst length may be the number of times data is continuously read or written by decrementing or incrementing the address from the initial address. In an example embodiment, when the burst length is 8, the memory device **200** may perform a burst read operation or a burst write operation 8 times consecutively from the initial address in response to a clock signal CLK.

[0081] For example, the column address latch **250** may increment the received column address COL\_ADDR in burst mode. The column address latch **250** may apply a temporarily stored or incremented column address COL\_ADDR' to each of the plurality of column decoders **270\_1** to **270\_n**.

[0082] A column decoder activated by the bank control circuit **230**, among the plurality of column decoders **270\_1** to **270\_n**, may activate a sense amplifier corresponding to the bank address BANK\_ADDR and column address COL\_ADDR through the input/output gating circuit **290**.

[0083] The input/output gating circuit **290** may include circuits for gating input/output data. Also, the input/output gating circuit **290** may include data latches for storing codewords output from the plurality of bank arrays **310\_1** to **310\_n** and write drivers for writing data in the plurality of bank arrays **310\_1** to **310\_n**.



[0084] In an example embodiment, during a read operation, a codeword CW read from a selected bank array, among the plurality of bank arrays **310\_1** to **310\_n**, may be detected by a sense amplifier corresponding to the selected bank array and stored in the data latches of the input/output gating circuit **290**. Also, the codeword CW stored in the data latches may be subjected to ECC decoding by the on-die ECC circuit **350** to be provided as data DTA to the data input/output buffer **320**. The data input/output buffer **320** may generate a data signal DQ based on data DTA and provide the data signal DQ to the memory controller **100** together with a strobe signal DQS.

[0085] In an example embodiment, data DTA to be written in a selected bank array, among the plurality of bank arrays **310\_1** to **310\_n**, may be received as a data signal DQ by the data input/output buffer **320** during a write operation. The data input/output buffer **320** may convert the data signal DQ into data DTA and provide the data DTA to the on-die ECC circuit **350**. The on-die ECC circuit **350** may generate parity bits or parity data based on the data DTA and provide a codeword CW, including the data DTA and the parity bits, to the input/output gating circuit **290**. The input/output gating circuit **290** may write the codeword CW in the normal cell area NA and the parity cell area PA of the selected bank array.

[0086] The data input/output buffer **320** may convert a data signal DQ into data DTA and provide the data DTA to the on-die ECC circuit **350** during a write operation. The data input/output buffer **320** may convert the data DTA, provided from the on-die ECC circuit **350**, into a data signal DQ during a read operation.

[0087] The on-die ECC circuit **350** may perform ECC encoding on the data DTA during a write operation. The on-die ECC circuit **350** may perform ECC decoding on the codeword CW during a read operation.

[0088] According to an example embodiment, the on-die ECC circuit **350** may perform an error correction operation on a data set input or output through the memory device **200** based on the codeword CW and the parity check matrix (or H-matrix) during a read operation, and may generate a first flag regarding a result of performing the error correction operation. The first flag may indicate non-uncorrectable error (non-UE) or uncorrectable error (UE). The non-UE indicates there is no error in the data set or that there was an error but the error has been corrected, and the UE indicates an uncorrectable error has occurred in the data set.

[0089] In an example embodiment, the on-die ECC circuits **350** included in each of the plurality of memory devices **200p** and **200d\_1** to **200d\_n** may be corresponding devices. For example, the on-die ECC circuit **350** provided in each of the plurality of memory devices **200p** and **200d\_1** to **200d\_n** may be the same model or product.

[0090] The control logic circuit **210** may control the operation of the memory device **200**. For example, the control logic circuit **210** may generate control signals such that the memory device **200** performs a write operation, a read operation, a normal refresh operation, and a hammer refresh operation. The control logic circuit **210** may include a command decoder **211**, decoding a command CMD received from the memory controller **100**, and a mode register set (MRS) **212** setting an operation mode of the memory device **200**.

[0091] The command decoder **211** may decode the command CMD to generate internal command signals such as an internal active signal IACT, an internal precharge signal IPRE, an internal read signal IRD, and an internal write signal IWR. Also, the command decoder **211** may decode a chip select signal and a command/address signal to generate control signals corresponding to the command CMD.

[0092] According to an example embodiment, a new read command (hereinafter, referred to as, for example, a read retry command) may be defined to request the first flag generated by the on-die ECC circuit **350** together with a data set. When a read retry command is received from the memory controller **100**, the command decoder **211** may decode the read retry command and control the operation of the memory device **200** to provide the first flag along with the data set.

[0093] The mode register set **212** may set the operation mode of the memory device **200**.

According to an example embodiment, the mode register set **212** may include a mode register setting the operation mode of the memory device **200** to a first mode or a second mode. The first mode may be a mode in which the memory device **200** operates to provide only a data set in response to a normal read command, and the second mode may be a mode in which the memory device **200** operates to provide the first flag generated by the on-die ECC circuit **350** together with the data set in response to a normal read command.

[0094] For example, when a first value is set in the mode register, the memory device **200** may operate in the first mode. Also, when a second value different from the first value is set in the mode register, the memory device **200** may operate in the second mode. The memory controller **100** may change a set value of the mode register through, for example, a mode register write (MRW) command to change the operation mode of the memory device **200**.

[0095] According to an example embodiment, the memory device **200** may include a pin allocated to output the first flag regarding a result of the error correction operation performed by the on-die ECC circuit **350**. The memory device **200** may apply a signal corresponding to the first flag generated by the on-die ECC circuit **350** to the pin, and the memory controller **100** may obtain the first flag based on the signal applied to the pin.

[0096] Alternatively, according to an example embodiment, the memory device **200** may extend the burst length. For example, the memory device **200** may select a single burst length from among a plurality of burst lengths and operate based on the selected burst length. Accordingly, the memory device **200** may expand the burst length by changing the burst length to a second burst length, which is relatively larger than the first burst length, during an operation based on the first burst length selected from among the plurality of burst lengths. The memory device **200** may provide the first flag, generated by the on-die ECC circuit **350**, to the memory controller **100** using burst(s) added based on the extended burst length.

[0097] FIG. **5** is a block diagram of a memory controller according to an example embodiment. The memory controller **100** of FIG. **5** may correspond to, for example, the memory controller **100** of FIG. **1**.

[0098] Referring to FIG. **5**, the memory controller **100** may include a processor **110**, a system ECC circuit **120**, a refresh logic (i.e., refresh logic circuit) **130**, a host interface **140**, a scheduler (i.e., scheduler circuit) **150**, and a memory interface **160**, all of which are connected to each other via a bus.

[0099] The processor **110** may control the overall operation of the memory controller **100**. For example, the processor **110** may control each of the system ECC circuit **120**, the refresh logic **130**, the host interface **140**, the scheduler **150**, and the memory interface **160**.

[0100] The refresh logic **130** may generate a command to perform a normal refresh operation on the plurality of wordlines of the memory device **200**. For example, the refresh logic **130** may generate an auto-refresh command for sequentially refreshing the plurality of wordlines depending on a refresh period.

[0101] The host interface **140** may interface with the host **15**.

[0102] The scheduler **150** may manage the scheduling and transmission of sequences of commands generated within the memory controller **100**.

[0103] The memory interface **160** may interface with the memory module **2000**.

[0104] The system ECC circuit **120** may store a second data set in the parity memory device **200p**. The second data set may include parity bits for a plurality of first data sets, respectively stored in the plurality of data memory devices **200d\_1** to **200d\_n**. According to an example embodiment, one of the parity bits included in the second data set and the data bits corresponding to the parity bit may have different bit positions. To this end, the system ECC circuit **120** may map each of the data bits and a corresponding parity bit to have different bit positions. For example, the system ECC circuit **120** may map data bits, respectively selected from the plurality of first data sets, and corresponding parity bits to have different DQs (or DQ positions). Also, the system ECC circuit

**120** may map data bits, respectively selected from the plurality of first data sets, and corresponding parity bits to have different burst orders (or BL positions). Also, the system ECC circuit **120** may map data bits, respectively selected from the plurality of first data sets, and corresponding parity bits to have different BLs and DQ positions.

[0105] The system ECC circuit **120** may perform a system ECC operation based on the plurality of first data sets and the second data set obtained from the memory module **2000** through a read operation. The system ECC operation may include various operations for correcting errors included in one of the plurality of first data sets based on the second data set.

[0106] For example, when the plurality of first data sets and the second data set are received from the plurality of data memory devices **200d\_1** to **200d\_n** and the parity memory device **200p**, the system ECC circuit **120** may perform a parity check on the received data.

[0107] According to an example embodiment, the parity bits included in the second data set may be even XOR parities. The system ECC circuit **120** may compare a result of the XOR operation of data bits with the parity bit to perform a parity check. For example, when the result of the XOR operation matches the parity bit, it may be determined that the parity bit and the data bits mapped to the parity bit do not include error bits. When the result of the XOR operation does not match the parity bit, it may be determined that there is an error bit among the parity bit and the data bits mapped to the parity bit.

[0108] For example, the parity check may be performed to determine whether errors have occurred in the plurality of first data sets and the second data set. Also, the parity check may be performed to identify candidate error bit positions in each of the plurality of first data sets and the second data set.

[0109] However, even when the parity check is performed, it is not specified which of the plurality of first data sets and the second data set includes actual error bits. Therefore, the system ECC circuit **120** may specify an error data set including error bits by performing the following operations.

[0110] For example, when first flags corresponding to each of the plurality of first data sets and the second data set are obtained, the system ECC circuit **120** may determine an error data set including error bits, among the plurality of first data sets and the second data set, based on the obtained first flags.

[0111] According to an example embodiment, the system ECC circuit **120** may determine the error data set when the first flags have specific values. Also, according to an example embodiment, the system ECC circuit **120** may perform an additional iteration operation to determine the error data set, which will be described later, even when the first flags do not have the specific values.

According to an example embodiment, the system ECC circuit **120** may include a parity check matrix used by the on-die ECC circuit **350** of the memory device **200**, and may perform an iteration operation using the parity check matrix. Alternatively, according to an example embodiment, the system ECC circuit **120** may perform an iteration operation through the plurality of data memory devices **200d\_1** to **200d\_n** and the parity memory device **200p**. This will be described in detail later.

[0112] As described above, when the candidate error bit positions and the error data set are identified, the system ECC circuit **120** may perform an error correction operation by flipping bits corresponding to the candidate error bit positions among the bits included in the error data set.

[0113] Hereinafter, the system ECC operations according to example embodiments will be described in detail with reference to FIGS. **6** to **11D**.

[0114] FIG. **6** is a flowchart illustrating a method of operating a memory system according to an example embodiment. The memory controller **100** and the memory module **2000** of FIG. **6** may correspond to the memory controller **100** and the memory module **2000** included in the memory system **20** of FIG. **1**, respectively. In FIG. **6**, a plurality of data memory devices **200d\_1** to **200d\_n** included in the memory module **2000** are storing the above-described plurality of first data sets,

respectively. Also, the parity memory device **200p** included in the memory module **2000** is storing the above-described second data set.

[0115] Referring to FIG. 6, in operation **S605**, the memory controller **100** may transmit a read command to the memory module **2000**. The read command may be a normal read command, but example embodiments are not limited thereto. For example, when data is requested by a host **15**, the memory controller **100** may transmit a read command to each of the plurality of data memory devices **200d\_1** to **200d\_n** and the parity memory device **200p**.

[0116] In operation **S610**, each of the plurality of data memory devices **200d\_1** to **200d\_n** and the parity memory device **200p** may perform a read operation in response to the read command and provide a plurality of first data sets Data and a second data set Parity to the memory controller **100**.

[0117] In operation **S615**, the memory controller **100** may perform a parity check on the plurality of first data sets and the second data set. When no error is detected through the parity check in operation **S615**, the flow proceeds to operation **S620** in which the memory controller **100** may transmit the plurality of first data sets to the host **15**.

[0118] When candidate error bit positions are identified through the parity check in operation **S615**, the memory controller **100** may obtain first flags Flag\_1, corresponding to each of the plurality of first data sets and the second data set, from the plurality of data memory devices **200d\_1** to **200d\_n** and the parity memory device **200p**, respectively. The first flag Flag\_1 may indicate uncorrectable error (UE) or uncorrectable error (non-UE). However, example embodiments are not limited thereto, and the first flag Flag\_1 may also include no error (NE) indicating that no error is detected, correctable error (CE) indicating that an error has been corrected, or uncorrectable error (UE). NE and CE may correspond to non-UE.

[0119] According to an example embodiment, in operation **S625**, the memory controller **100** may transmit a read retry command to the plurality of data memory devices **200d\_1** to **200d\_n** and the parity memory device **200p**. Accordingly, in operation **S630**, each of the plurality of data memory devices **200d\_1** to **200d\_n** and the parity memory device **200p** may provide the first flag Flag\_1 to the memory controller **100** in response to the read retry command.

[0120] According to an example embodiment, each of the plurality of data memory devices **200d\_1** to **200d\_n** and the parity memory device **200p** may include the mode register described in FIG. 4. The memory controller **100** may change a set value of the mode register to a second value using a mode register write command. Then, the memory controller **100** may transmit a normal read command to each of the plurality of data memory devices **200d\_1** to **200d\_n** and the parity memory device **200p** to obtain the first flags Flag\_1.

[0121] In operation **S635**, the memory controller **100** may attempt to identify an error data set based on the first flags Flag\_1. For example, when only one of the first flags Flag\_1 is UE, the memory controller **100** may identify a data set corresponding to the UE, among the plurality of first data sets and the second data set, as the error data set.

[0122] When an error data set is identified in operation **S635**, the flow proceeds to operation **S640** in which the memory controller **100** may flip bits corresponding to candidate error bit positions among bits included in the error data set. Accordingly, errors that have occurred in one of the plurality of first data sets and the second data set may be corrected.

[0123] According to an example embodiment, one of the plurality of first data sets may be an error data set. In operation **S645**, the memory controller **100** may transmit the remaining first data sets, excluding the error data set, and an error data set, in which bits corresponding to the candidate error bit positions are flipped, to the host **15**.

[0124] When only one of the first flags Flag\_1 is not UE in operation **S635**, for example, when the error data set is not identified, the flow proceeds to operation **S650** in which the memory controller **100** may perform an iteration operation. For example, when all of the first flags Flag\_1 indicate non-UE in operation **S635**, the flow proceeds to operation **S650** in which the memory controller **100** may perform an iteration operation. The iteration operation will be described in detail with

reference to FIG. 8.

[0125] FIGS. 7A and 7B are diagrams illustrating a method of operating a memory system according to an example embodiment. FIG. 7A illustrates first data sets Data 1 and Data 2 stored in two data memory devices each having a prefetch size of 16 bits and a second data set parity stored in a parity memory device. FIG. 7B illustrates a state after an error correction operation of the memory system 20 is performed on the data sets of FIG. 7A.

[0126] FIGS. 7A and 7B illustrate an example in which each data bit and a corresponding parity bit are mapped to have different DQ positions, as illustrated in FIG. 2C. In FIGS. 7A and 7B, the operation of the memory system 20 will be described by way of an example in which three errors indicated by “X” occur in Data 1.

[0127] When Data 1, Data 2, and Parity as illustrated in FIG. 7A are received through a read operation, the memory controller 100 may perform a parity check on Data 1, Data 2, and Parity and identify candidate error bit positions.

[0128] Referring to FIG. 7A, errors are present in bit positions (BL0, DQ3), (BL1, DQ1), and (BL3, DQ2) among bit positions of Data 1. Therefore, the memory controller 100 may identify that errors have occurred in bit positions (BL0, DQ1), (BL1, DQ3), and (BL3, DQ0) (indicated by “!”), among the bit positions of Parity, through a parity check between the mapped data bits and parity bit.

[0129] Accordingly, the memory controller 100 may identify (BL0, DQ3) of Data 1 and (BL0, DQ2) of Data 2 mapped to (BL0, DQ1) of Parity, together with (BL0, DQ1) of Parity, as candidate error bit positions. Also, the memory controller 100 may identify (BL1, DQ1) of Data 1 and (BL1, DQ0) of Data 2 mapped to (BL1, DQ3) of Parity, together with (BL1, DQ3) of Parity, as candidate error bit positions. Also, the memory controller 100 may identify (BL3, DQ2) of Data 1 and (BL3, DQ1) of Data 2 mapped to (BL3, DQ0) of Parity, together with (BL3, DQ0) of Parity, as candidate error bit positions.

[0130] As a result, the memory controller 100 may identify the candidate error bit positions for Data 1 as (BL0, DQ3), (BL1, DQ1), and (BL3, DQ2), the candidate error bit positions for Data 2 as (BL0, DQ2), (BL1, DQ0), and (BL3, DQ1), and the candidate error bit positions for Parity as (BL0, DQ1), (BL1, DQ3), and (BL3, DQ0).

[0131] The candidate error bit positions have been identified, so that the memory controller 100 may obtain the first flags Flag\_1 corresponding to Data 1, Data 2, and Parity.

[0132] In an example embodiment, the two data memory devices corresponding to Data 1 and Data 2 and the parity memory device corresponding to Parity may each include the on-die ECC circuit 350 using a single error correction, double error detection (SECDED) Hamming code. As shown in FIG. 7A, there are three bits of error in Data 1, and the on-die ECC circuit 350 of the data memory device corresponding to Data 1 may therefore generate UE as the first flag Flag\_1. Also, there is no error in Data 2, so that the on-die ECC circuit 350 of the data memory device corresponding to Data 2 may generate non-UE as the first flag Flag\_1. Also, there is no error in Parity, so that the on-die ECC circuit 350 of the parity memory device may generate non-UE as the first flag Flag\_1.

[0133] According to an example embodiment, the memory controller 100 may obtain the first flag Flag\_1 generated from each memory device using a read retry command or a mode register.

[0134] The memory controller 100 may attempt to identify the error data set based on the obtained first flags Flag\_1. For example, when only one of the first flags Flag\_1 is UE, the memory controller 100 may identify the data set corresponding to the UE as an error data set among the plurality of first data sets and the second data set. Referring to FIG. 7A, only the first flag Flag\_1 corresponding to Data 1 is UE, so that the memory controller 100 may identify Data 1, a data set corresponding to UE, as an error data set.

[0135] Accordingly, the memory controller 100 may flip bits corresponding to the candidate error bit positions, among bits included in the error data set. In the example of FIG. 7A, the error data set is Data 1 and the candidate error bit positions for Data 1 are identified as (BL0, DQ3), (BL1, DQ1),

and (BL3, DQ2). The memory controller **100** may therefore flip bits corresponding to (BL0, DQ3), (BL1, DQ1), and (BL3, DQ2) among the bits of Data **1**. As a result, three error bits included in Data **1** may be corrected, as illustrated in FIG. 7B.

[0136] FIG. **8** is a flowchart illustrating a method of operating a memory system according to an example embodiment. The operations of FIG. **8** may be an example of the iteration operation in operation S**650** of FIG. **6**. As described above with reference to FIG. **6**, according to an example embodiment, the memory controller **100** may perform an iteration operation when first flags Flag\_1 are all non-UE.

[0137] Referring to FIG. **8**, in operation S**651**, the memory controller **100** may flip bits corresponding to candidate error bit positions in each of a plurality of first data sets and a second data set. Accordingly, a plurality of third data sets, respectively corresponding to the plurality of first data sets, and a fourth data set corresponding to the second data set may be generated.

[0138] In operation S**652**, the memory controller **100** may perform an error correction operation on each of a plurality of third data sets and a fourth data set using a parity check matrix or (H-matrix), and may generate a second flag Flag\_2 regarding the error correction operation performed for each of the plurality of third data sets and the fourth data set.

[0139] The second flag Flag\_2 may indicate no error (NE) correctable error (CE), or uncorrectable error (UE). NE may indicate no error is detected. CE may indicate that an error has been corrected. UE may indicate that an uncorrectable error has occurred.

[0140] According to an example embodiment, the parity check matrix used by the memory controller **100** to generate the second flag Flag\_2 may be the same as the parity check matrix used by an on-die ECC circuit **350** included in each of a plurality of memory devices **200p** and **200d\_1** to **200d n**.

[0141] In operation S**653**, the memory controller **100** may attempt to identify an error data set based on the second flags Flag\_2. For example, when only one of the second flags Flag\_2 corresponding to each of the plurality of third data sets and the fourth data set is NE, the memory controller **100** may identify a data set corresponding to NE, among the plurality of first data sets and the second data set, as an error data set. For example, the second flags Flag\_2 correspond to the plurality of third data sets and the fourth data set but the plurality of third data sets respectively correspond to the plurality of first data sets and the fourth data set corresponds to the second data set, so that a data set corresponding to NE, among the plurality of first data sets and the second data sets, may be identified.

[0142] When the error data set is identified in operation S**653**, the flow proceeds to operation S**654** in which the memory controller **100** may flip bits corresponding to candidate error bit positions, among bits included in the error data set. Accordingly, errors that have occurred in one of the plurality of first data sets and the second data set may be corrected.

[0143] According to an example embodiment, one of the plurality of first data sets may be an error data set. In operation S**655**, the memory controller **100** may transmit the remaining first data sets, excluding the error data set, and an error data set, in which the bits corresponding to the candidate error bit positions are flipped, to a host **15**.

[0144] When only one of the second flags Flag\_2 is not NE in operation S**653**, for example, when the error data set is not identified, the flow proceeds to operation S**656** in which the memory controller **100** may determine that an error has been detected in the plurality of first data sets and the second data set, but that the error was uncorrectable. Such a situation may be referred to as detected but uncorrectable error (DUE), but example embodiments are not limited thereto. For example, when two or more of the second flags Flag\_2 are NE in operation S**653**, the memory controller **100** may determine that DUE has occurred.

[0145] While an example has been provided in which the memory controller **100** performs the iteration operations in operations S**651** to S**656**, example embodiments are not limited thereto.

[0146] For example, when the memory controller **100** does not identify an error data set in

operation S635, the memory controller **100** may control the plurality of memory devices **200p** and **200d\_1** to **200d\_n** to flip the bits corresponding to the candidate error bit positions in each of the plurality of first data sets and the second data set, thereby generating a plurality of third data sets and a fourth data set.

[0147] Also, the memory controller **100** may control the plurality of memory devices **200p** and **200d\_1** to **200d\_n** to perform an error correction operation on each of the plurality of third data sets and the fourth data set, thereby generating second flags Flag\_2. Each of the second flags Flag\_2 may be generated by the on-die ECC circuit **350** included in each of the plurality of memory devices **200p** and **200d\_1** to **200d\_n**.

[0148] Also, the memory controller **100** may control the plurality of memory devices **200p** and **200d\_1** to **200d\_n** to transmit the generated second flag Flag\_2.

[0149] Accordingly, the memory controller **100** may perform an iteration operation by performing operations S653 to S656 based on the obtained second flags Flag\_2.

[0150] As described above, the memory system **20** may increase the likelihood of detecting an error data set through the iteration operation. Also, when an error data set is not identified, the memory system **20** may eliminate the likelihood of silent data corruption (SDC), in which data that should be corrected is not corrected while data that should not be corrected is corrected, by treating such a situation as DUE without performing further error correction operations.

[0151] FIGS. 9A to 9D are diagrams illustrating a method of operating a memory system according to an example embodiment. FIG. 9A illustrates a situation in which an on-die ECC circuit **350** of a data memory device corresponding to Data 1 fails to detect errors in Data 1 in the same situation as FIG. 7A. In the FIG. 9A, three errors in Data 1 are indicated by “X” and bit positions of Parity identified by the memory controller **100** through a parity check between the mapped data bits and parity bit are indicated by “!”. The on-die ECC circuit **350** is unable to detect 100% of errors included in a data set, so that such a situation may occur.

[0152] Regardless of the fact that the on-die ECC circuit **350** corresponding to Data 1 has failed to detect the error in Data 1, the memory controller **100** may identify candidate error bit positions through a parity check operation on Data 1, Data 2, and Parity (for example, operation S615 of FIG. 6). The candidate error bit positions are the same as those described in FIG. 7A.

[0153] Accordingly, the memory controller **100** may obtain the first flags Flag\_1 from each memory device and attempt to identify the error data set based on the first flags Flag\_1 (for example, operation S635 of FIG. 6).

[0154] However, all of the first flags Flag\_1 obtained by the memory controller **100** from each memory device indicate non-UE as illustrated in FIG. 9A, so that the error data set may not be identified and the memory controller **100** may perform an iteration operation (for example, operation S650 of FIG. 6).

[0155] For example, the memory controller **100** may flip bits corresponding to candidate error bit positions in each of Data 1, Data 2, and Parity to generate new data sets Data 3, Data 4, Parity' as illustrated in FIG. 9B (for example, operation S651 of FIG. 8).

[0156] Referring to FIG. 9B, Data 3 may be generated by flipping the bits corresponding to the candidate error bit positions in Data 1, so that errors present in Data 1 are no longer present in Data 3. There were no errors in Data 2 and Parity, so that errors indicated by “X” may occur in Data 4 and Parity' generated by flipping the bits corresponding to the candidate error bit positions.

[0157] The memory controller **100** may perform an error correction operation on each of Data 3, Data 4, and Parity' using a parity check matrix, such as the parity check matrix used in the memory devices storing Data 1, Data 2, and Parity, and may generate second flags Flag\_2 regarding the error correction operation (for example, operation S652 of FIG. 8).

[0158] FIG. 9C illustrates a second flags Flag\_2 generated for each of Data 3, Data 4, and Parity'. Referring to FIG. 9C, there is no error in Data 3, so that the second flag Flag\_2 for Data 3 may be NE. Also, there are errors in Data 4 and Parity' (indicated by “X”), so that the second flags Flag\_2

for Data 4 and Parity' may be CE or UE.

[0159] The memory controller **100** may attempt to identify the error data set based on the second flags Flag\_2 (for example, operation S653 of FIG. 8). For example, when only one of the second flags Flag\_2 is NE, the memory controller **100** may identify the data set corresponding to NE, among Data 1, Data 2, and Parity, as the error data set. In the example of FIG. 9C, only the second flag Flag\_2 corresponding to Data 3 is NE, so that the memory controller **100** may identify Data 1, corresponding to Data 3, as the error data set among Data 1, Data 2, and Parity.

[0160] Accordingly, the memory controller **100** may flip the bits corresponding to the candidate error bit positions, among the bits included in the error data set. In the example of FIG. 9A, the error data set is Data 1 and the candidate error bit positions for Data 1 are identified as (BL0, DQ3), (BL1, DQ1), and (BL3, DQ2), so that the memory controller **100** may flip bits corresponding to (BL0, DQ3), (BL1, DQ1), and (BL3, DQ2) in Data 1, among the bits of Data 1. As a result, the three error bits included in Data 1 may be corrected, as illustrated in FIG. 9D.

[0161] FIGS. 10A and 10B are diagrams illustrating limitations of an error correction operation during related parity mapping. Referring to FIGS. 10A and 10B, in the case of a related technique, a parity bit for data bits having the same bit position in a first data set Data 1 and another first data set Data 2 may be stored in the same bit position of a second data set Parity.

[0162] FIG. 10A illustrates a situation in which errors occur in Data 1 as illustrated in FIG. 9A, but the on-die ECC circuit **350** of the data memory device corresponding to Data 1 fails to detect the errors and generates the first flag Flag\_1 as NE. There are no errors in Data 2 and Parity, so that the corresponding on-die ECC circuits **350** may also generate the first flag Flag\_1 as NE. In the FIG. 10A, three errors in Data 1 are indicated by "X" and bit positions of Parity identified through a parity check between the mapped data bits and parity bit are indicated by "!".

[0163] The first flags Flag\_1 are all NE or non-UE, so that the candidate error bit positions may be identified through a parity check operation on Data 1, Data 2, and Parity, while the error data set is not identifiable.

[0164] In the case of a related technique, by flipping the bits corresponding to the candidate error bit positions among the bits included in Data 1, Data 2, and Parity, new data sets Data 3, Data 4, and Parity' are generated, as illustrated in FIG. 10B. Referring to FIG. 10B, Data 3 may be generated by flipping the bits corresponding to the candidate error bit positions in Data 1, so that errors present in Data 1 are no longer present in Data 3. There were no errors in Data 2 and Parity, so that errors indicated by "X" may occur in Data 4 and Parity' generated by flipping the bits corresponding to the candidate error bit positions. However, in the case of a related technique, even if the second flags Flag\_2 are obtained for each of Data 3, Data 4, and Parity', the error data set is not identifiable.

[0165] For example, the on-die ECC circuit **350** of the data memory device corresponding to Data 1, the on-die ECC circuit **350** of the data memory device corresponding to Data 2, and the on-die ECC circuit **350** of the parity memory device may be the same product. For example, the on-die ECC circuit **350** of the data memory device corresponding to Data 1 failed to detect the errors in (BL0, DQ3), (BL1, DQ1), and (BL3, DQ2), which means that the on-die ECC circuits **350** of other memory devices also fail to detect errors occurring in (BL0, DQ3), (BL1, DQ1), and (BL3, DQ2).

[0166] Accordingly, even when the second flags Flag\_2 are obtained using Data 3, Data 4, and Parity' as illustrated in FIG. 10B, the second flags Flag\_2 all become NE and the error data set is no longer identifiable.

[0167] In example embodiments, parity mapping is performed to have different bit positions, so that even when the same on-die ECC circuit **350** or the same parity check matrix is used, an additional error data set may be detected through an iteration operation.

[0168] FIGS. 11A to 11D are diagrams illustrating a method of operating a memory system according to an example embodiment. Errors in data are indicated by "X" and bit positions of Parity identified through a parity check between the mapped data bits and parity bit are indicated



by “!”. FIGS. 11A and 11B illustrate the same situations as illustrated in FIGS. 9A and 9B, respectively. Therefore, the duplicate descriptions are omitted to avoid redundancy.

[0169] After the iteration operation is started and new data sets Data 3, Data 4, Parity' are generated, the memory controller 100 may generate second flags Flag\_2 for Data 3, Data 4, and Parity'.

[0170] FIG. 11C illustrates a second flag Flag\_2 generated for each of Data 3, Data 4, and Parity'. Referring to FIG. 11C, there is no error in Data 3, so that the second flag Flag\_2 for Data 3 may be NE.

[0171] There are errors in Data 4 and Parity', so that the second flags Flag\_2 for Data 4 and Parity' may be CE or UE, as described above. However, as described above, the likelihood of the on-die ECC circuit 350 detecting an error is not 100%, which means that the second flag for at least one of Data 4 and Parity' may be NE. FIG. 11C illustrates a case in which the second flag Flag\_2 for Data 4 is NE.

[0172] When the second flag Flag\_2 for Data 4 is NE, and Data 2 is also identified as an error data set, flipping candidate error bit positions for Data 2 may result in silent data corruption (SDC) in which data that should not be corrected is erroneously corrected. Accordingly, according to an example embodiment, when two or more of the second flags Flag\_2 are NE, the memory controller 100 may determine that an error has been detected but is not correctable for the plurality of first data sets and the second data set. The operation of correcting data that should not be corrected is not performed, so that SDC may not occur.

[0173] For example, referring to the example of FIG. 11C, the second flags Flag\_2 for both Data 3 and Data 4 are NE, indicating that the memory controller 100 may interpret the situation as an error being detected but uncorrectable for Data 1, Data 2, and Parity. As illustrated in FIG. 11D, the error correction operation may no longer be performed for Data 1, Data 2, and Parity.

[0174] FIG. 12 is a block diagram of an electronic system according to an example embodiment. While a description has been provided for an example in which the system ECC circuit 120 is included in the memory controller 100, example embodiments are not limited thereto. For example, the system ECC circuit 120 performing the above-described system ECC operations may be implemented independently of the memory controller 100.

[0175] Referring to FIG. 12, an electronic system 10' may have the same configuration as the electronic system 10 of FIG. 1, except that the system ECC circuit 120 is included in a host 15 and external to the memory controller 100. Therefore, redundant descriptions are omitted. Various system ECC operations, except for data exchange with the memory module 2000 performed by the above-described memory controller 100, may be performed by the system ECC circuit 120 included in the host 15.

[0176] A technology for correcting all errors for a single data chip among data chips constituting a cache line size is referred to as chipkill or single device data correction (SDDC). Generally, two parity memory chips per cache line size are required for implementation of chipkill technique. However, according to various example embodiments, a high-performance chipkill technique may be implemented using a single system ECC chip.

[0177] While aspects of example embodiments have been particularly shown and described, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the following claims.

## Claims

1. A memory system comprising: a plurality of first memory devices; a second memory device; and a memory controller configured to control the plurality of first memory devices and the second memory device, wherein each of the plurality of first memory devices is configured to store a first data set comprising a first set of data bits in different bit positions, each of the bit positions

corresponding to a burst order and a DQ, wherein the second memory device is configured to store a second data set comprising parity bits for a plurality of first data sets, respectively stored in the plurality of first memory devices, and wherein a second set of data bits corresponding to a first parity bit of the parity bits are included in the plurality of first data sets, respectively, and at least two data bits of the second set of data bits have different bit positions.

**2.** The memory system of claim 1, wherein the first parity bit has a value based on an exclusive OR (XOR) operation performed on the second set of data bits.

**3.** The memory system of claim 1, wherein the at least two data bits have different burst orders or different DQs within corresponding at least two first data sets.

**4.** The memory system of claim 1, wherein each of the plurality of first memory devices and the second memory device is further configured to perform a first error correction operation on a data sets input or output through a corresponding memory device using a parity check matrix, and generate a first flag regarding a result of the first error correction operation, and wherein the memory controller is further configured to perform a parity check on the plurality of first data sets and the second data set received from the plurality of first memory devices and the second memory device, and obtain the first flag from the plurality of first memory devices and the second memory device to obtain first flags based on candidate error bit positions for each of the plurality of first data sets and the second data set being identified through the parity check.

**5.** The memory system of claim 4, wherein the memory controller is further configured to transmit the plurality of first data sets to a host based on the candidate error bit positions not being identified through the parity check.

**6.** The memory system of claim 4, wherein the memory controller is further configured to obtain the first flags based on a new read command or based on a setting value in a mode register included in each of the plurality of first memory devices and the second memory device, wherein each of the plurality of first memory devices and the second memory device is configured to provide, based on the new read command, the first flag regarding the result of the first error correction operation and a corresponding data set, and wherein each of the plurality of first memory devices and the second memory device is configured to selectively operate, based on the setting value, in a first mode, in which only the corresponding data set is provided based on a normal read command, and a second mode in which both the first flag regarding the result of the first error correction operation and the corresponding data set are provided based on the normal read command.

**7.** The memory system of claim 4, wherein each of the plurality of first memory devices and the second memory device is further configured to provide the first flag to the memory controller through a pin assigned to the first flag or by using an extended burst length.

**8.** The memory system of claim 4, wherein the first flag indicates non-uncorrectable error (Non-UE) based on determining there is no error or that the error has been corrected, and uncorrectable error (UE) based on determining an uncorrectable error has occurred, and wherein the memory controller is further configured to identify an error data set corresponding to the first flag indicating the UE, among the plurality of first data sets and the second data set, based on only one of the first flags indicating the UE, and flip bits corresponding to the candidate error bit positions included in the identified error data set to generate a flipped error data set.

**9.** The memory system of claim 8, wherein the memory controller is further configured to, based on the error data set being included in the plurality of first data sets, transmit remaining first data sets, excluding the error data set, and the flipped error data set to a host.

**10.** The memory system of claim 4, wherein the memory controller is further configured to: generate a plurality of third data sets, respectively corresponding to the plurality of first data sets, and a fourth data set corresponding to the second data set, by flipping bits corresponding to the candidate error bit positions among bits included in each of the plurality of first data sets and the second data set, based on all of the first flags indicating non-UE; and perform a second error correction operation on each of the plurality of third data sets and the fourth data set using the

parity check matrix and generate a second flag indicating a result of the second error correction operation for each of the plurality of third data sets and the fourth data set.

**11.** The memory system of claim 10, wherein the second flag indicates no error (NE) based on determining there is no error, correctable error (CE) based on determining the error has been corrected, and uncorrectable error (UE) based on determining the uncorrectable error has occurred, and wherein the memory controller is further configured to: identify an error data set corresponding to the NE among the plurality of first data sets and the second data set based on only one of second flags, respectively corresponding to the plurality of third data sets and the fourth data set, indicating the NE; and flip bits corresponding to the candidate error bit positions among bits included in the identified error data set to generate a flipped error data set.

**12.** The memory system of claim 11, wherein the memory controller is further configured to, based on the error data set being included in the plurality of first data sets, transmit remaining first data sets, excluding the error data set, and the flipped error data set to a host.

**13.** The memory system of claim 11, wherein the memory controller is further configured to determine errors were detected but uncorrectable for the plurality of first data sets and the second data set, based on two or more of the second flags indicating the NE.

**14.** A memory module comprising: a plurality of first memory devices; and a second memory device, wherein each of the plurality of first memory devices is configured to store a first data set comprising a first set of data bits having bit positions, each of the bit positions corresponding to a burst order and a DQ, wherein the second memory device is configured to store a second data set comprising first parity bits for a plurality of first data sets, respectively stored in the plurality of first memory devices, and wherein a second set of data bits corresponding to one parity bit of the first parity bits are included in the plurality of first data sets, respectively, and at least two data bits of the second set of data bits have different bit positions.

**15.** The memory module of claim 14, wherein the at least two data bits have different burst orders or different DQs within corresponding at least two first data sets.

**16.** The memory module of claim 14, wherein each of the plurality of first memory devices and the second memory device comprise a bank array comprising a plurality of memory cells, and an on-die error correction code (ECC) circuit, wherein the bank array of each of the plurality of first memory devices and the second memory device comprises a normal region, in which a data set is stored, and a parity region, in which second parity bits for the data set stored in the normal region are stored, and wherein the on-die ECC circuit of each of the plurality of first memory devices and the second memory device is further configured to perform an error correction operation on a data set input or output through a corresponding memory device based on a codeword comprising the second parity bits and a parity check matrix, and generate a first flag regarding a result of the error correction operation.

**17.** The memory module of claim 16, wherein each of the plurality of first memory devices and the second memory device comprises mode register configured to store a setting value indicating either a first mode or a second mode, and wherein each of the plurality of first memory devices and the second memory device is further configured to provide only a corresponding data set based on a normal read command while the setting value indicates the first mode, and provide both the first flag regarding the result of the error correction operation and the corresponding data set based on the normal read command while the setting value indicates the second mode.

**18.** The memory module of claim 16, wherein each of the plurality of first memory devices and the second memory device is further configured to provide the first flag regarding the result of the error correction operation to an external entity through a pin assigned to the first flag or by using an extended burst length.

**19.** A method of operating a memory controller to control a memory module including a plurality of first memory devices and a second memory device, the method comprising: storing a plurality of first data sets in the plurality of first memory devices, respectively; and storing a second data set,

comprising parity bits for the plurality of first data sets, in the second memory device, wherein each of the plurality of first data sets comprises a first set of data bits having bit positions, each of the bit positions corresponding to a burst order and a DQ, and wherein a second set of data bits corresponding to a first parity bit of the parity bits are included in the plurality of first data sets, respectively, and bit positions of at least two data bits of the second set of data bits are different from each other.

**20.** The method of claim 19, comprising: receiving the plurality of first data sets and the second data set from the plurality of first memory devices and the second memory device; performing a parity check on the plurality of first data sets and the second data set to identify candidate error bit positions for each of the plurality of first data sets and the second data set; obtaining first flags regarding results of error correction operations, performed on each of the plurality of first data sets and the second data set, by receiving a first flag from each of the plurality of first memory devices and the second memory device; and identifying a data set comprising error bits, among the plurality of first data sets and the second data set, based on the first flags.

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